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Security Infrastructure as Code

for Entra ID and Graph API

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Agenda

Scope for this Security Infrastructure as Code session..

1. Azure Resource Manager

and..

2. Bicep
3. Terraform
4. Bicep Graph
5. Terraform AzureAD
6. Terraform Graph

1. Bicep vs. Terraform — Key Differences

- Bicep is Azure-native, compiles to ARM, day-zero support for new Azure services
- Terraform uses HCL with provider model — supports multi-cloud and third-party integrations
- Bicep is stateless (ARM manages state); Terraform requires explicit state file management
- Terraform has a larger public module ecosystem; Bicep's AVM (Azure Verified Modules) is growing
- Both now support Microsoft Graph / Entra ID resources via extensions or providers

2. Bicep Graph Provider

- Built on Bicep extensibility — deploys Graph resources alongside Azure resources via ARM
- Supported resources: **Applications, Service Principals, Groups, Federated Identity Credentials, App Role Assignments, OAuth2 Permission Grants**
- Users are read-only — can reference existing users but cannot create or modify
- No support for application passwords (secrets) — only key credentials
- Role-assignable groups cannot be deployed — use DeploymentScript workaround
- No what-if preview, no deployment stacks, no verbose output support

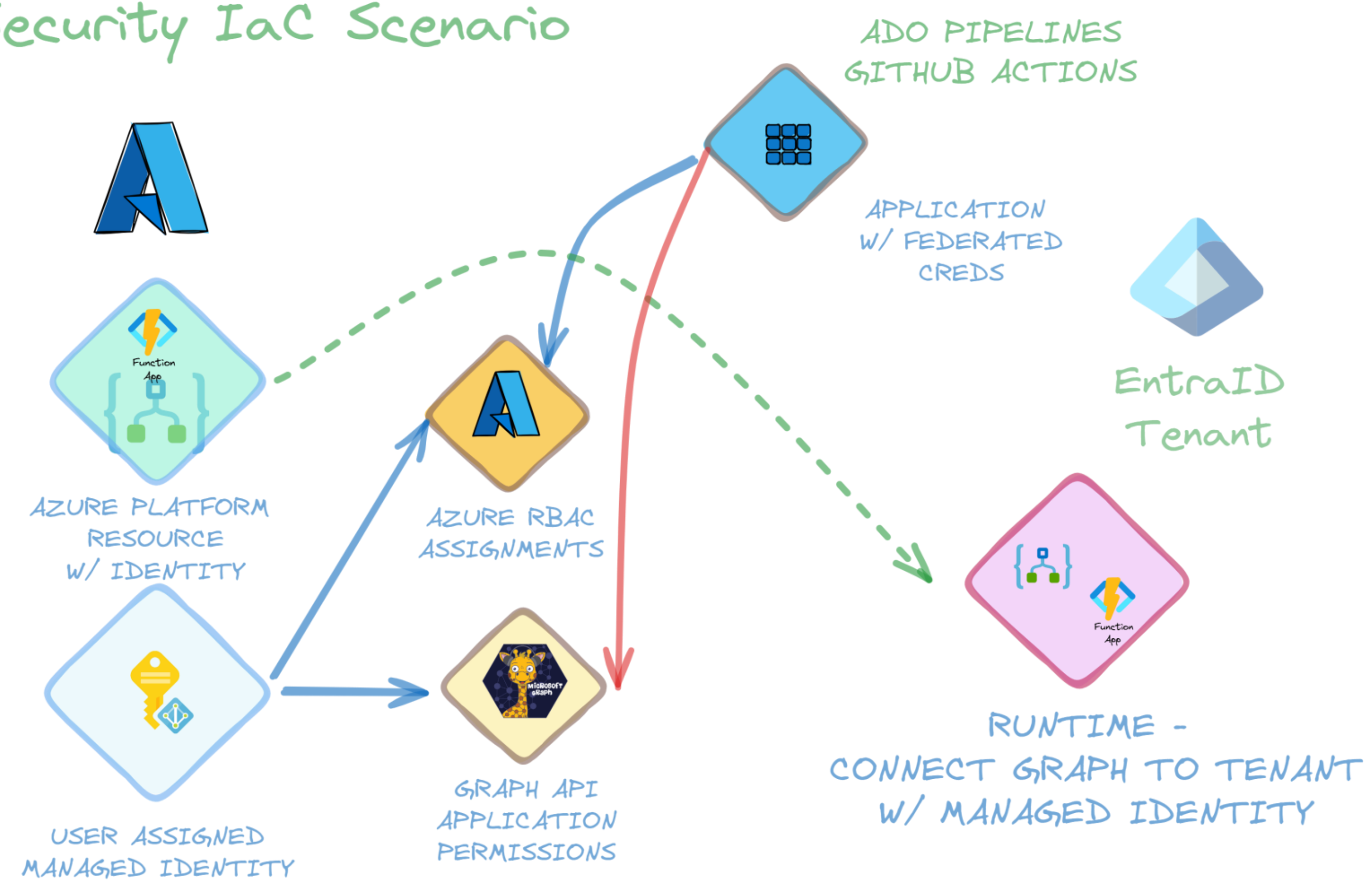
3. Terraform AzureAD Provider

- Maintained by HashiCorp — the most mature Terraform provider for Entra ID
- Manages users, groups, applications, service principals, and conditional access policies
- Migrated from legacy Azure AD Graph API to Microsoft Graph in v2.0
- Current version 3.x — rebuilt on a new Microsoft Graph SDK for better performance
- Purpose-built resources with strongly typed schemas and full lifecycle management
- Supports 56 resources and 21 data sources — the broadest Entra ID coverage in Terraform

4. Terraform MSGraph Provider

- Developed by Microsoft — currently in preview, covers all Microsoft Graph endpoints
- Uses generic resource types (msgraph_resource) with direct Graph API JSON bodies
- Extends beyond Entra ID to M365 APIs: SharePoint, Intune, Privileged Identity Management
- vs. AzureAD: generic approach vs. purpose-built resources; broader scope but less type safety
- vs. Bicep Graph: Terraform state-based lifecycle with drift detection; Bicep is stateless via ARM
- Currently in preview (v0.3.0) — requires users to construct Graph API URLs and JSON payloads

Security IaC Scenario



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5. Security Discussion and Scenarios

- Is this wise?
- Too much privileges?
- Difference between Global Admin doing this as scriptet operations vs. this?
- To Pipeline/Action or not?
- Local only?

Questions?

You can reach me at

- @JanVidarElven
- https://github.com/janvidarelven/security-infrastructure-as-code

Please evaluate this session in the App.

THANK YOU

Are there any questions?

Next session 11:10 – 12:10

Defending the Defender: Avoiding Misconfigurations in Defender XDR, Thomas Verheyden, Room 4-5

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